TLP:AMBER

HTOC

HEALTHCARE THREAT OPERATIONS CENTER

Next_Obs_Daily Batch Process — Daily Full Report Build

Version 1.0
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1. Purpose

This SOP documents the automated **Next_Obs_Daily** batch process that consolidates partner-level NOI prediction outputs into a single daily file for downstream reporting and distribution. The process runs unattended via Task Scheduler and provides run-level logging for operations support.

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2. Scope

This SOP applies to HTOC analysts and engineers responsible for operating, monitoring, and troubleshooting the `Next_Obs_Daily` scheduled task. It covers the batch launcher, Python processing logic, logging, validation, and failure handling.

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3. Prerequisites

3.1 Runtime and Access

Requirement	Notes
Python 3.13 via `py -3.13`	Required by batch guard clause in `.bat`
pandas, openpyxl	Installed on each run by the batch file (`pip install --user`)
Access to `Z:\HTOC\Data_Analytics\`	Required for input and output paths

Requirement	Notes
Task Scheduler access on F.R.E.D	Required to verify task history and trigger reruns

Example dependency install command:

```
pip install pandas openpyxl
```

3.2 Input Data Availability

The process expects partner files to already exist for the execution date under:

`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\{PartnerName}\{YYYYMMDD}.csv`

If no files exist for the current date, the script exits cleanly with `No data to save.`

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4. Process Components

Component	Path	Role
Batch launcher	`scripts/batch-processing-script/Next_Obs_Daily/next_observed_daily_reports.bat`	Validates Python runtime, installs dependencies, runs script, writes logs

Component	Path	Role
Main script	`scripts/batch-processing-script/Next_Obs_Daily/src/main.py`	Loads today's partner CSVs, adds metadata columns, writes consolidated CSV
Structured log	`scripts/batch-processing-script/Next_Obs_Daily/run_log.json`	JSON history of timestamp + output line per run
Raw output log	`scripts/batch-processing-script/Next_Obs_Daily/output.log`	Full pip + script stdout/stderr stream

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5. Execution Flow

Step	Action	Expected Result
1	Task Scheduler triggers `next_observed_daily_reports.bat` at 8:15 AM	Batch starts in script directory
2	Batch verifies `py -3.13` availability	Continues if present; exits with error if missing
3	Batch runs pip install for `pandas openpyxl`	Dependencies ensured for current user profile
4	Batch executes `src/main.py`	Script scans partner folders and builds `daily_search` dataframe

Step	Action	Expected Result
5	Script writes `full_daily_report_{YYYYMMDD}.csv` if not already present	Consolidated output file exists in `Full Daily Reports`
6	Batch appends run entry to `run_log.json` and full output to `output.log`	Run is auditable

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6. Python Script Logic (`src/main.py`)

6.1 Key constants

Constant	Value
`DATA_PATH`	`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions`
`SAVE_PATH`	`Z:\HTOC\Data_Analytics\Data\OpDiv_Predictions\Full Daily Reports`
`EXCLUDE_FOLDERS`	`automation scripts`, `Logs`, `LogsBackup`, `Full Daily Reports`

6.2 Data processing behavior

Scans all subfolders under `DATA_PATH`.

Filters files to **today only** (`YYYYMMDD.csv` naming pattern).

Skips excluded folders to prevent recursion into output/log paths.

Adds columns:

`Partner` (folder name)

`FileDate` (date extracted from filename)

Concatenates all matching files into one dataframe.

Writes output only if file does not already exist for today.

6.3 Idempotency guard

If `full_daily_report_{YYYYMMDD}.csv` already exists, the script prints `File already exists: ...` and does not overwrite it.

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7. Batch Launcher Behavior (`next_observed_daily_reports.bat`)

Forces execution in script directory (`cd /d "%~dp0"`).

Enforces Python 3.13 usage (`py -3.13`) to avoid older pip/runtime issues.

Sets user-scoped package paths (`PYTHONUSERBASE`, `PYTHONPATH`).

Installs dependencies with retries and trusted hosts.

Runs Python script and captures output.

Appends structured run output to `run_log.json` via PowerShell JSON append logic.

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8. Run Success Criteria

A successful run should satisfy all of the following:

Task Scheduler result code is success (`0x0`).

`run_log.json` has a new entry with today's timestamp.

Entry output is either:

`Saved to ...full_daily_report_{YYYYMMDD}.csv`, or

`File already exists: ...full_daily_report_{YYYYMMDD}.csv.`

Output file exists and is readable at:

[Z:\HTOC\Data Analytics\Data\OpDiv Predictions\Full Daily Reports\full daily report {YYYYMMDD}.csv].

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9. Troubleshooting

Symptom	Likely Cause	Resolution
`[ERROR] Python 3.13 not found` in `output.log`	Runtime missing on host	Install Python 3.13 and confirm `py -3.13` works from a new shell
`No CSV files found for today.` then `No data to save.`	Upstream partner files not present yet	Verify upstream NOI process completed and date-stamped files exist
`Skipping <file>` errors	One or more CSV files unreadable/corrupt	Inspect and fix malformed CSV in the partner folder
`File already exists: ...`	Re-run for same date	Expected idempotent behavior; no action needed
Task failed but no clear message in scheduler	Output only in log files	Review `output.log` and latest object in `run_log.json`

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10. Operator Decision Points

If no data exists by 8:15 AM, verify upstream forecast process before rerunning this batch.

If run fails due to environment/runtime, fix Python or package issues first, then rerun task manually.

If output exists but appears incomplete, validate partner source folders and rerun only after correcting source data.

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11. Related Documents

`SOP_Daily_Reports_Consolidation.md` — combined notebook + script process documentation

`SOP_NextObservedIndicator_Forecasting.md` — upstream forecast generation process

GitHub repository: [jaytlinaskew-OIS/HTOC](https://*github.com/jaytlinaskew-OIS/HTOC*)